

Mamadou Mbengue

+1 613-276-7783

90 University Private, Ottawa, Ontario, K1N 0E5
mmben104@uottawa.ca

SUMMARY OF QUALIFICATIONS

- Fourth year financial mathematics and economics student at the University of Ottawa
- **Bilingual**, fluent in **French** and **English**
- Familiar with **Regression** and **Value estimation**, **Classification** and **Supervised learning**, **Clustering** and **Unsupervised learning** in **Machine learning**
- Solid foundation in **stochastic calculus**, with **research experience** in **martingales**, **Brownian motion**, **Ito calculus**, and their **applications** to **financial modeling** (e.g., Black-Scholes, Greeks)
- Good knowledge of **Microsoft office**, **Canva**, **Microsoft Excel** and **PowerPoint**
- Experienced in **data analysis** using **R Studio**, **Python** and **SQL**
- Experienced in **data entry** using **TM1**
- Worked in various projects implying the use of **financial modeling** and **statistical analysis**
- Experienced with the use of **CRMs** (Customer Relationship Management)
- Completed **SQL for finance professionals**, and **EXCEL Managing and Analyzing Data** Certifications on LinkedIn Learning
- Completed **Python for Machine learning and Data Science Masterclass** on Udemy
- Applied machine learning to portfolio optimization by building a Random Forest model to predict Sharpe ratios and efficiently approximate the efficient frontier

EDUCATION

Honors Bachelor of Financial Mathematics and Economics (2021-Present)

University of Ottawa, Ottawa, Ontario

- Expected graduation date: December 2025
- Listed on Dean's honors list (Fall 2021-Winter 2022-Winter 2024-Winter 2025)
- Recipient of a Merit Scholarship (Fall 2021- Winter 2022-Winter 2024-Winter 2025)
- Tuition fee exemption scholarship (French immersion)

WORK EXPERIENCE

MATH RESEARCH ASSISTANT

May 2025- August 2025

University of Ottawa, Ontario

- Delivered biweekly presentations on advanced stochastic calculus topics, including martingales, Brownian motion, Wiener integrals, Ito's lemma, and stochastic differential equations
- Conducted independent research on the application of stochastic processes in finance
- Developed Python simulations to model financial instruments and pricing techniques, including the Black-Scholes model, Greeks analysis, and jump diffusion processes

MERCHANDISE FINANCIAL PLANNING INTERN

May 2024- August 2024

Giant Tiger, Ottawa, Ontario

- Supported in reviewing store plans across 221 classes and conducted clean-up to realign with compensation strategy
- Supported planning team with various tasks
- Data entry through TM1, Power Bi and AS400
- Compiled reports as required
- Created financial reports through Excel
- Trained on proprietary ERP system
- Participated to business meetings

AGENT- SUPPORT INTERN

September 2023 – December 2023

International Market development, University of Ottawa, Ottawa, Ontario

- Made data analysis on different market research projects
- Assisted in various projects as needed
- Assisted with administrative duties and made use of a CRM to perform various tasks such as logging interactions, updating contact information, and tracking communication history

EXAM PROCTOR

January 2023 – Present

Academic support, University of Ottawa, Ottawa, Ontario

- Proctored exams maintaining academic integrity
- Addressed student inquiries and maintained high level of confidentiality
- Utilized various assistive technologies such as Jaws, word, and Q to accommodate students with visual impairments

NOTETAKER ASSISTANT

September 2022 – Present

Academic support, University of Ottawa, Ottawa, Ontario

- Providing comprehensive notes for disabled students, ensuring accuracy and accessibility
- Organizing various notes through OneDrive files
- Receiving positive feedback for organized and timely notetaking

RESIDENCE ADVISOR (RA)

August 2022 – Present

Housing services, University of Ottawa, Ottawa, Ontario

- Organizing community-building events

- Contributing to increased resident engagement
- Documenting daily tasks using a CRM